



Lebanese cannabis oil as a potential treatment for acute myeloid leukemia: In vitro and in vivo evaluations

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Ethnopharmacological relevance: The *Cannabis sativa* L. ssp. *indica* (Lam.) plant has been historically utilized as a natural herbal remedy for the treatment of several ailments. In Lebanon, cannabis extracts have long been traditionally used to treat arthritis, diabetes, and cancer.

Aim of the study: The current study aims to investigate the anti-cancer properties of Lebanese cannabis oil extract (COE) on acute myeloid leukemia using WEHI-3 cells, and a WEHI-3-induced leukemia mouse model.

Materials and methods: WEHI-3 cells were treated with increasing concentrations of COE to determine the IC₅₀ after 24, 48 and 72-h post treatment. Flow cytometry was utilized to identify the mode of cell death. Western blot assay was performed to assess apoptotic marker proteins. *In vivo* model was established by inoculating WEHI-3 cells in BALB/c mice, and treatment commencing 10 days post-inoculation and continued for a duration of 3 weeks.

Results: COE exhibited significant cytotoxicity with IC₅₀ of 7.76, 3.82, and 3.34 µg/mL at 24, 48, and 72 h respectively post-treatment. COE treatment caused an induction of apoptosis through an inhibition of the MAPK/ERK pathway and triggering a caspase-dependent apoptosis via the extrinsic and intrinsic modes independent of ROS production. Animals treated with COE exhibited a significantly higher survival rate, reduction in spleen weight as well as white blood cells count.

Conclusion: COE exhibited a potent anti-cancer activity against AML cells, both *in vitro* and *in vivo*. These findings emphasize the potential application of COE as a chemotherapeutic adjuvant in treatment of acute myeloid leukemia.

1. Introduction

Leukemia refers to a diverse group of blood cancers originating from the irregular growth of white blood cells (WBCs). This proliferation originates from the dysregulated growth of hematopoietic stem cells residing within the bone marrow, which are responsible for the generation of these transforming leukocytes (Du et al., 2022). In 2020, leukemia ranked as the 15th most common cancer globally and the 11th leading cause of cancer-related deaths. Amid children under 15 years old, leukemia constitutes approximately 30% of all cancers, making it the most prevalent cancer type in this age group. It also accounts for the highest proportion of fatalities (Namayandeh et al., 2020; Du et al., 2022). There are several types of leukemia, categorized primarily by

their pace of progression, either rapid (acute) or gradual (chronic), and whether they originate in myeloid or lymphoid cells. Originating in the bone marrow, acute myeloid leukemia (AML) rapidly moves into the bloodstream and can disseminate to various body regions such as lymph nodes, liver, spleen, central nervous system and even the testicles. AML poses significant challenges in clinical management, necessitating a multi-faceted approach for effective treatment (Amjad et al., 2024). Therapeutic modalities for AML encompass a spectrum of strategies tailored to the patient's specific disease characteristics and overall health (Fleischmann et al., 2021). Chemotherapy stands as the primary modality employed to address AML (Stubbins et al., 2022), which involves employing antimetabolites, topoisomerase inhibitors, and anti-microtubular agents either singly or in various combinations. (Nygren, 2001; Schirmacher, 2019a; Amjad et al., 2024). Although

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Abbreviations

7-AAD	7-amino-actinomycin D
AML:	Acute Myeloid Leukemia
(Annexin V-PE)	Annexin V-Phycoerythrin
BSA	Bovine serum albumin
CBC	Cannabichromene
CBD	Cannabidiol
CBN	Cannabinol
COE	Cannabis Oil Extract
DCFDA	dichlorofluorescein diacetate
DMSO	Dimethyl sulfoxide
ERK	Extracellular signal-regulated kinases
FBS	Fetal Bovine Serum

H ₂ O ₂	Hydrogen Peroxide
HRP	Horseradish peroxidase
IC ₅₀	Half-maximal inhibitory concentration
LPS	Lipopolysaccharide
MAPK	Mitogen-activated protein kinases
PAGE	Polyacrylamide gel electrophoresis
PBS	Phosphate buffer solution
RIPA	Radio immunoprecipitation assay buffer
ROS	Reactive oxygen species
RPMI	Roswell Park Memorial Institute
SDS	Sodium dodecyl sulphate
TBST	Tris-buffered saline with Tween 20
THC	Tetrahydrocannabinol
WST	water soluble tetrazolium

chemotherapy sounds promising at first, it is still important to note that these drugs are still considered, to this day, as among the most harmful and toxic medications prescribed for therapeutic purposes (Anand et al., 2022). Therefore, to avoid the toxic side effects of chemotherapy, researchers have turned to plant-based medicine. This approach has involved the use of numerous traditional ethnic plants, which have shown promising results in treating various cancers. (Garcia-Oliveira et al., 2021; Gao et al., 2023, 2024).

Cannabis sativa L. (*C. sativa*), a herbaceous plant in the Cannabaceae family, has a longstanding historical presence as a natural herbal remedy and serves as the most significant source of phytocannabinoids (Atakan, 2012; Dariš et al., 2019). Cannabis and the endocannabinoid system have gained a lot of attention in recent years, especially since it was legalized for medical purposes in numerous countries such as the US, Canada, Portugal, Switzerland, and Argentina (Aguilar et al., 2018). In April 2020, the Lebanese parliament approved a draft law to legalize cannabis cultivation for medical and industrial purposes (Lebanese Official Gazette; issue 23; 2020).

Cannabis has been used medicinally for various purposes throughout history. Arab practitioners have used cannabis leaves for treating epilepsy, inflammation, nausea, fever, and pain (Lozano, 2001). Al-Intaqui's 17th-century pharmacopeia lists cannabis's euphoric and sedative effects and prescriptions for somatic diseases (Robinson et al., 2020). German botanist Rumphius discusses cannabis's potential for treating gonorrhoea, constipation, asthma, and poisoning (Joni et al., 2023). The world's earliest pharmacopeia, the pen-ts'ao ching, reports on the use of cannabis as medicine by ancient Chinese. It was based on oral traditions dating back to Emperor Shen-Nung's reign and prescribed for conditions like rheumatism, intestinal constipation, female reproductive system abnormalities, malaria, and more (Mack and Joy, 2000). Moreover, medical cannabis has been utilized in southern Asia for around 1000 years BC, with a variety of uses including analgesic, anticonvulsant, anti-inflammatory, antibiotic, antiparasitic, antispasmodic, digestive, antitumorogenic, appetite stimulant, and diuretic (Lozano, 2001; Zuardi, 2006; Pertwee, 2015; Shirah et al., 2023).

The Cannabis plant contains over 400 different chemical compounds, whereby over 100 of these are the 21-carbon terpenophenolic cannabinoids (Abrams, 2022). It contains a combination of various phytochemicals, including cannabinoids such as cannabidiol (CBD) and Δ^9 -tetrahydrocannabinol (THC) as well as monoterpenes (α -pinene, camphene, β -myrcene, and D-limonene) and sesquiterpenes (β -caryophyllene, α -bergamotene, α -humulene, humulene epoxide II, and caryophyllene oxide) (Legault and Pichette, 2007; Irrera et al., 2021). The importance of evaluating the anti-cancer activities of cannabis oil resides in its unique phytochemical combination, which may exhibit an entourage effect resulting in enhanced synergistic activity rather than using one chemical component alone (Russo, 2019; Koltai and Namdar, 2020). Cannabinoids have been demonstrated to be beneficial in the

treatment of various forms of cancers (Carracedo et al., 2006; Dariš et al., 2019; Tomko et al., 2020; Irrera et al., 2021). Preclinical studies revealed that THC and CBD exert their anticancer effects through multiple mechanisms involving inhibition of migration and/or invasion (Milian et al., 2020), apoptosis (Lombard et al., 2005; Fisher et al., 2016; Tomko et al., 2020), autophagy (Salazar et al., 2009; Shrivastava et al., 2011), and regulation of the tumour microenvironment (Pagano et al., 2021; Sun et al., 2023). One of the most significant and studied phytocannabinoids found in cannabis species is CBD, which was found to be beneficial in the treatment of melanoma, leukemia (Hinz and Ramer, 2022), breast, prostate, colon, cervical, ovarian, and endometrial malignancies (Kalenderoglou et al., 2017; Romano et al., 2014; Tomko et al., 2020). THC, the primary psychoactive compound found in cannabis, displayed anti-cancer activity against various types of cancer cell lines, including non-small cell lung cancer (NSCLC) cells such as A549, H460, H1792, and SW-1573 (Preet et al., 2008; Milian et al., 2020), triple negative breast cancer cells such as MDA-MB-231 (Cafarrel et al., 2006), glioma cells represented by the C6 cell line (Goncharov et al., 2005; Blázquez et al., 2008), and the Jurkat leukemic cell line (Jia et al., 2006).

Cannabichromene (CBC) is regarded as one of the minor phytocannabinoids produced by *Cannabis sativa* L. plants. CBC exhibits a range of therapeutic properties such as anti-cancer, anti-inflammatory and anti-microbial actions (Taura et al., 2007). Other minor components of COE such as β -myrcene exhibits analgesic (Rao et al., 1990), anti-inflammatory (Rufino et al., 2015), antibacterial (Inoue et al., 2004), and antimutagenic effects (Bai and Tang, 2020). β -caryophyllene was also reported to possess anti-inflammatory, anti-cancer, and cytoprotective effects on the gastric mucosa (Brenneisen, 2007; Francomano et al., 2019).

In Lebanon, cannabis oil has long been employed in traditional medicine to address a range of health issues, including cancer (Fakhry et al., 2021). Unlike conventional chemotherapy, which often causes harmful side effects, and can lead to resistance to multiple drugs, cannabis oil offers promise as a safer alternative (McKallip et al., 2006; Olivás-Aguirre et al., 2019; Tomko et al., 2020). To the best of our knowledge, there is no existing literature addressing the entourage effect of cannabis against leukemia, particularly acute myeloid leukemia (AML). Consequently, the present study is designed to investigate the anti-leukemic effect of a Lebanese cannabis oil extract on the WEHI-3 cell line, as well as the WEHI-3-induced leukemia mouse model.

2. Materials and Methods

2.1. Chemicals and reagents

Dimethyl sulphoxide (DMSO), Roswell Park Memorial Institute (RPMI) (with 20 mM HEPES/L-glutamine/without sodium bicarbonate/

liquid, sterile-filtered), supplemented with: 10% heat-inactivated sterile and filtered fetal bovine serum (FBS) purchased from Sigma Aldrich, Missouri, USA, alongside 1% of a 10,000 U/mL Penicillin-streptomycin (Pen/Strep) from Gibco, Germany. Tris buffered saline (TBS10X), Laemli buffer, ECL, Bio-Rad protein assay kits, β -mercaptoethanol, goat anti-rabbit secondary antibody (horseradish peroxidase (HRP) conjugate), and goat anti-mouse secondary antibody (HRP Conjugate) were purchased from Bio-Rad (Hercules, CA, USA). Spectra multicolour broad range protein ladder was purchased from ThermoFisher, UK. Annexin V-Phycoerythrin (Annexin V-PE) and 7-AAD (7-amino-actinomycin D) Guava Nexin apoptosis detection reagent kit was purchased from Luminex, Austin, Texas, U.S.A. All other chemicals used in the experiments were of analytical grade.

2.2. Plant collection and extraction

The Drug Enforcement Office in Zahle, Beqaa Governorate provided plant samples of a Lebanese cannabis strain. These samples were collected from Yammoune, Beqaa Valley, with specific coordinates (Yammoune: 34°07'46.4" North, 36°01'40.8" East; altitude, 1375 $\pm$ 10 m) in October 2021. Following collection, the plant material was securely stored on campus within a protected storage facility. A specimen (ID, 2021–002) was archived at the Department of Natural Sciences at the Lebanese American University. A 10 g sample of the plant material underwent ethanol extraction for 48 h. The resulting extract was filtered and concentrated at 45 °C under reduced pressure, producing 1.12 g of cannabis oil extract (COE).

2.3. Tested organisms

All *in vivo* experiments were performed on white male adult BALB/c mice (10–12 weeks old) bred in the animal facility at the Lebanese American University. Mice, weighing 27–29 g were housed under optimum conditions of temperature set at 22 $\pm$ 2 °C, humidity at 50 $\pm$ 5% and an alternating cycle of light and dark. The animals were supplied with a standard laboratory chow diet and water. All experimental protocols were approved by the Animal Ethical Committee of the Lebanese American University (LAU.ACUC.SOP.MM1.22/March/2023) which complies with the Guide for the Care and Use of Laboratory Animals (Carracedo et al., 2006).

2.4. Cell lines and culture

WEHI-3 cell lines were purchased from the American Type Culture Collection (ATCC, USA-TIB68). These cell lines cultured in RPMI supplemented with 10% heat-inactivated fetal bovine serum (FBS), 100 μ g/mL streptomycin and 100 U/mL penicillin. Cells were maintained in a humidified atmosphere containing 5% CO₂ at 37 °C.

2.5. Cytotoxicity and cell survival assays

The proliferation of WEHI-3 cell lines was tested using WST-1 assay. Cells were plated in 96-welled plates at a concentration of 5 $\times$ 10⁵ cell/mL. Afterwards, cells were treated with increasing concentrations (2.5, 5, 10, 20, 40, and 80 μ g/mL) of the crude oil dissolved in DMSO for 24 h, 48 h, and 72 h. At the end of the treatment period, WST-1 reagent (Roche Applied Science, Penzberg, Germany) was added to the cells and incubated in a humidified incubator at 37 °C and 5% CO₂ atmosphere for 2 h. In metabolically active cells, mitochondrial dehydrogenases break WST-1, a water-soluble tetrazolium salt. The Multiskan FC microplate ELISA reader was used to measure the amount of formazan generated at 450 nm (Thermo fisher Scientific, Rockford, IL, USA). Cells were plated in triplicates in three independent experiments.

2.6. Cell death analysis using flow cytometry

Annexin V-Phycoerythrin (Annexin V-PE) and 7-AAD (7-amino-actinomycin D) staining (Guava Nexin Reagent Kit, Luminex, Austin, Texas, U.S.A), were used to analyze the type of cell death at 24 h post-treatment. Briefly, WEHI-3 cells were seeded in a 6-well plate (2.5 $\times$ 10⁵ cells/mL) and treated with IC₅₀ and 2xIC₅₀ of COE and incubated at 37 °C/5% CO₂ for 24 h. After 24 h, cells were collected, and 10 μ L of the mix was then added to 10 μ L Trypan blue to count the cells under the microscope (Nikon Eclipse TE300). A final concentration of 500 cells/ μ L was obtained and adjusted per condition and the cell suspension was centrifuged at 1500 rpm for 5 min at 4 °C. The supernatant was discarded, and the pellet was suspended in 100 μ L Guava Nexin Reagent buffer, provided by the kit. Each sample was pipetted onto a 96-well plate and incubated in the dark for 10 min. Cells were then analyzed using the Guava® easyCyte 8HT Benchtop Flow Cytometer (Millipore, Luminex, USA). Annexin V/7-AAD data was measured on FL1-H versus FL2-H scatter plot. Cells treated with 1% DMSO were used as a negative control and Cytarabine was used as an apoptotic control after 24 h treatment.

In single cell gating, gates were set based on forward scatter (FSC) and side scatter (SSC) plots. This procedure is critical for excluding debris, ensuring that the analysis focuses solely on single cells. Afterwards, a dot plot is generated to illustrate the relationship between Annexin V-PE and 7-AAD staining.

Establishing four crucial gates is vital for accurately differentiating various cell populations in this analysis. These gates include viable cells (Annexin V-/7AAD-), early apoptotic cells (Annexin V+/7AAD-), late apoptotic cells (Annexin V+/7AAD+), and necrotic cells (Annexin V-/7AAD+). These techniques are essential for accurately distinguishing cell states, particularly in studies of cell viability and apoptosis.

2.7. Cell lysate preparation and protein quantification for western blot

WEHI-3 cells (5 $\times$ 10⁵ cell/ml) were seeded onto 6-well plates in the presence of different concentrations (IC₅₀ and 2xIC₅₀) and treated for 24h. The cells were collected on ice and washed with PBS and lysed with RIPA buffer whereby they were centrifuged at 15,000 rpm for 15 min at 4 °C. The supernatants (containing the proteins) were collected and then stored at –80 °C for western blotting. Total proteins of cell lysate were determined using Bradford protein assay kit.

2.8. Sodium dodecyl Sulphate-PolyAcrylamide gel electrophoresis and western blot analysis

WEHI-3 cell lysate protein samples were heated at 100 °C for 5 min. Then, using SDS-PAGE under reducing conditions, equal protein quantities of each sample (35 μ g) were loaded and resolved using 5% stacking gel, 10–15% separating gel, and 1X running buffer (0.3% Tris Base, 1.4% glycine, 20% SDS, pH = 8.3) at 80 V for 30 min, and then at 120 V for 2 h. Following that, proteins were transferred to PVDF membrane (Pall Corporation, Ann Arbor, USA) and blocked with blocking solution (5% BSA) for 1 h. Subsequently, the membranes were incubated at 4 °C overnight with the following primary antibodies: Actin (Internal Reference; 1:1000; rabbit; Cell Signalling Technology #4970) Bax (1:2000; rabbit; Invitrogen MA5-32031), Bcl-2(1:2000; rabbit; Invitrogen PA5-27094), Cytochrome C(1:2000; rabbit; Invitrogen PA5-86022), PARP-1(1:1000; rabbit; Cell Signalling Technology #9542), Pro-Caspase 8(1:1000; mouse; Invitrogen MA5-15914), Pro-Caspase-3 (1:2000, rabbit, PA5-96077), Cleaved-Caspase-3(1:500; rabbit; Invitrogen PA5-1146871), Cleaved-Caspase 8(1:500; rabbit; Invitrogen PA5-20118), Erk(1:1000; rabbit; Invitrogen MA5-15134), and p-ERK (1:000; rabbit; Invitrogen MA5-36265). Primary antibodies were then washed away with TBST for 30 min, and then the membranes were treated with horseradish peroxidase (HRP)-coupled secondary goat anti-mouse (1:5000; goat; Bio-Rad 170-5047) and goat anti-rabbit(1:5000;

goat; Bio-Rad 170-5046) antibodies in blocking solution for 1 h before being washed with TBST afterwards. Finally, using the chemiluminescence ECL kit, each protein was detected (Abcam plc, Cambridge, UK). Detection of proteins was performed using the chemiluminescence ECL kit and blot images were obtained and analyzed (Hodroj et al., 2018) with the image lab Software (Chemidoc imaging instrument, Bio-Rad).

2.9. ROS production

To detect and investigate the production or reduction of reactive oxygen species (ROS) in COE treated cells, the ROS DCFDA/H2DCFDA-cellular Assay Kit (Abcam, Cambridge, USA) was utilized. WEHI-3 cells were plated into a 96-well dark plate at 100,000 cells/well using phenol red free media and treated with IC₅₀ and 2xIC₅₀. After 20 h, hydrogen peroxide (200 μ M) was utilized as a positive control and administered for 4 h. After 24 h (24 h for COE treated and 4 h for H₂O₂ treated), cells were washed once with PBS and stained with DCFDA of 20 μ M concentration. DCFDA (100 μ L/well of 20 μ M) was kept on the cells for 45 min in the dark and then removed and 100 μ L of RPMI phenol red free media was added over the wells. The plate was scanned using the Varioskan™ LUX Multimode microplate reader (Thermo Fisher) to measure the fluorescence at an excitation/emission of 485/535 nm.

2.10. Establishment of the leukemic mice model

The total number of WEHI-3 cells was counted and dispensed into individual syringes at a concentration of 1×10^6 cells/ml in PBS. The mice were then divided into 5 groups with 8–9 animals each to ensure equal representation. The cells were then intraperitoneally injected (I. P.) into the mice, followed by a waiting period of 10 days before conducting a blood smear using Wright's stain (Himedia, Mumbai, India). Leukemia induction, group distribution, and treatment establishment were based on similar protocols reported in literature (Yang et al., 2006; Salim et al., 2014).

2.11. In vivo assays

The mice were divided randomly into 5 groups each of 9 mice. Group I was the control and contained normal animals. Group II served as a WEHI-3 control and was injected intraperitoneal (i.p) with the vehicle consisting of: PBS: ethanol: Tween 80 (18:1:1). Groups III, IV, V were injected i.p with 5, 10, 20 mg/kg of COE in PBS: ethanol: Tween 80 (18:1:1) respectively. The treatment was given for 3 weeks three times/week. Mice were sacrificed post treatment, the spleens were removed onto ice, weighed, and placed in pre-chilled tubes containing formaldehyde. The survival rate was calculated and plotted.

2.12. Statistical analysis

The data and results were presented as mean $\pm$ SEM, which were obtained from three separate experiments conducted independently. The Shapiro-Wilk test was used to confirm for normality. A one-way ANOVA was performed followed by a Bonferroni post hoc test for multiple comparisons, and statistical significance was considered when $p < 0.05$. GraphPad Prism 9 software was utilized for statistical analysis and data visualization.

3. Results

3.1. COE chemical composition (GCMS)

The cannabis oil extract was prepared and analyzed using GC-MS in one of our previous studies. (Shebaby et al., 2021). Briefly, CBD had the highest percentage of 59.1%, accompanied by THC constituting 20.2% of our extract. The remaining cannabinoids compromised a lower

percentage: CBN (3.6%) and CBC (2.2%). The identified monoterpenes constitute 4.12% of the extract and include β -Myrcene (1.94%), α -Pinene (1.01%), and D-Limonene (0.51%). Lastly, sesquiterpenes formed around 5.89% of the extract, whereby β -Caryophyllene, α -Bergamotene, and α -Humulene constitute 1.78%, 1.74% and 0.53% respectively (Shebaby et al., 2021).

3.2. Cytotoxicity assay (WST-1 assay)

The cytotoxic effect of the COE was studied on WEHI-3 cancer cells for 24, 48, and 72h using the WST-1 cell proliferation assay. Multiple concentrations (2.5, 5, 10, 20, 40, 80 μ g/mL) of COE were used. Results (Fig. 1A) showed a time and dose-dependent inhibitory effect on cell proliferation. The obtained IC₅₀ values were 7.76 ± 0.66 μ g/mL for 24h, 3.82 ± 1.03 μ g/mL for 48 h, and 3.34 μ g/mL ± 0.33 for 72 h. Moreover, the cytotoxic effects of COE are also apparent as demonstrated by phase-contrast microscopy (Fig. 1B). The morphological changes in the WEHI-3 cell line are detected at 20 μ g/mL and 40 μ g/mL of COE treatment after 24, 48 and 72h. In the control group, most cells displayed a normal morphology. However, following the administration of cannabis oil, a considerable amount of cells exhibited cytoplasmic protrusions, apoptosis, blebbing, and growth inhibition.

3.3. Flow cytometry

Annexin V/7-AAD staining was used to determine the type of cell death in WEHI-3 cells following 24 h of incubation with COE. The results (Fig. 2) showed that cell death was primarily due to apoptosis, as evidenced by a significant decrease in viable cells at 24 h post-treatment with COE (22.53%) and cytarabine (23.26%) compared to the control cells. Additionally, there was a significant increase in the percentage of late apoptotic cells at 24 h post-treatment with COE and cytarabine, with minimal contribution from necrosis.

3.4. Western blots

The effect of COE on the expression of key apoptotic and autophagic proteins in WEHI-3 cells was measured by Western blot analysis at 24h post COE treatment. Cells were treated with IC₅₀ (7.76 μ g/mL) and 2xIC₅₀ (15.52 μ g/mL), and Western blot results were analyzed using ImageLab software and normalized accordingly. Results (Fig. 3) have shown that COE treatment at IC₅₀ and 2xIC₅₀ had a significant effect on the expression of all pro-apoptotic markers represented below, at 24 h post-treatment. The pro-apoptotic protein BAX expression level was significantly increased compared to the control, thus indicating a pro-apoptotic mechanism via an increased Bax/Bcl2 ratio at 24h. This increase was also accompanied by a significant increase in the expression levels of cytochrome c, and cleaved caspase 3, alongside a decrease of Pro-Caspase 3, and Pro-PARP, at 24 h confirming a programmed mode of cell death. Furthermore, a significant reduction in procaspase 8, alongside a significant increase in cleaved caspase 8 expression was observed at the 24 h mark, indicating the supplemental activation of the extrinsic pathway as an additional factor in initiating cell death. Additionally, we examined the role of the MAP Kinase pathway in cell death by assessing the phosphorylation level of ERK. According to the findings, P-ERK was also observed to be significantly inhibited after 24 h post COE treatment.

3.5. ROS production

WEHI-3 cells were treated with IC₅₀ and 2xIC₅₀ of COE for 24 h, and then stained by 20 μ M DCFDA for 45 min in the dark. Upon ROS production, the DCFDA produced a fluorescent product. The measured fluorescence was therefore proportional to the produced ROS. Cells treated with COE showed a significant decrease in ROS production (Fig. 4) when compared to control group as well as the positive control

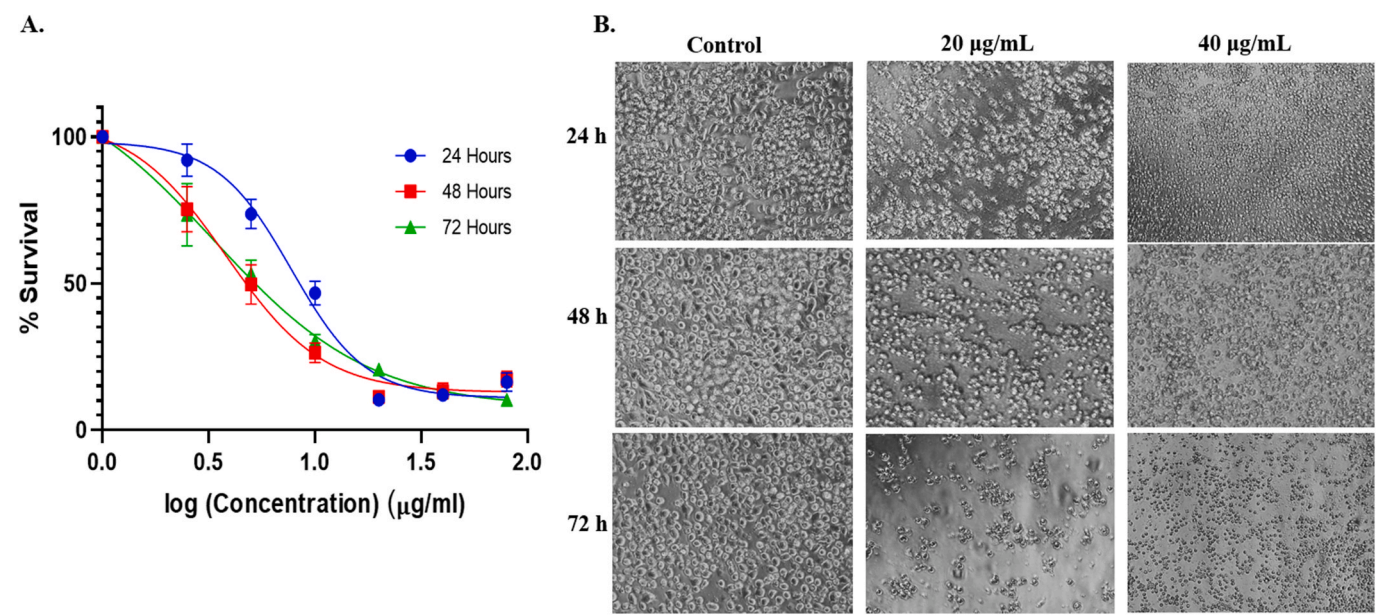


Fig. 1. Cytotoxicity dose response of cannabis oil in WEHI-3 cells. (A) Treatment was administered using 2-fold dilutions (80, 40, 20, 10, 5, 2.5 µg/mL). Cells were incubated for 24h, 48h, and 72h. Data are expressed as % survival as a function of log (concentration). The data points represent the mean value ± standard error of the mean (SEM), derived from three independent experiments run in triplicate (n = 3). (B) Phase contrast microscopic images were captured at 0, 24, 48, and 72 h (Magnification: 20x).

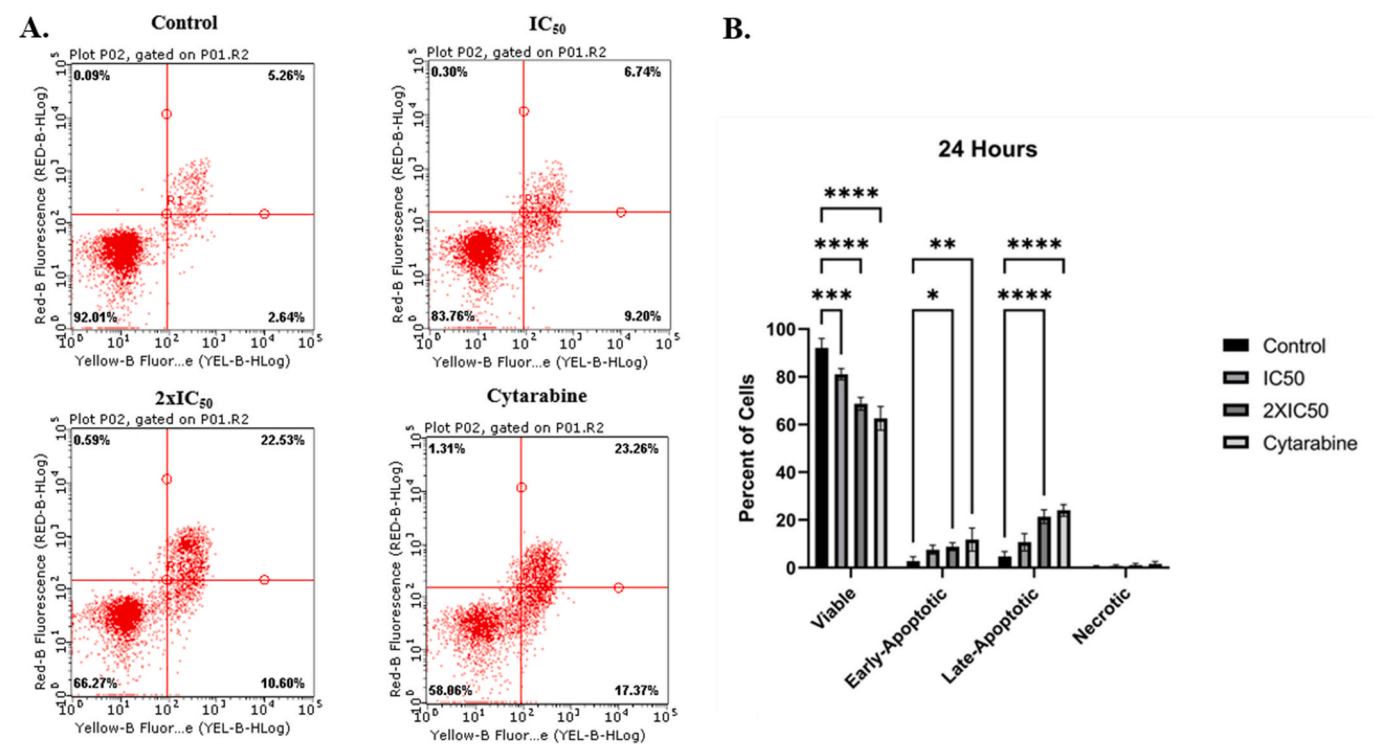


Fig. 2. Flow cytometric analysis of cell death mediated by COE. WEHI-3 cells were treated with the COE and analyzed 24h post-treatment. (A) Representative contour plots. (B) Bar graph representing percent viable, early-apoptotic, late-apoptotic, and necrotic cells at 24h. Cytarabine was used as a positive control. Bars denote mean ± SEM calculated from 3 independent experiments. *p < 0.05, **p < 0.01, ***p < 0.001 and ****p < 0.0001 versus control.

(Hydrogen Peroxide).

3.6. In vivo study

3.6.1. Establishing the model

Prior to commencing the 3-week treatment experiment, the validity of the model was confirmed by sacrificing mice, collecting their blood,

and analyzing the spleen weights of the WEHI-3 injected groups in comparison to the control group. A significant increase in spleen weight was observed in the WEHI-3 group when compared to the control group (Fig. S1A.). A blood smear was performed using wright’s staining, and an increase in the number of abnormally shaped white blood cells was observed (Fig. S1B.).

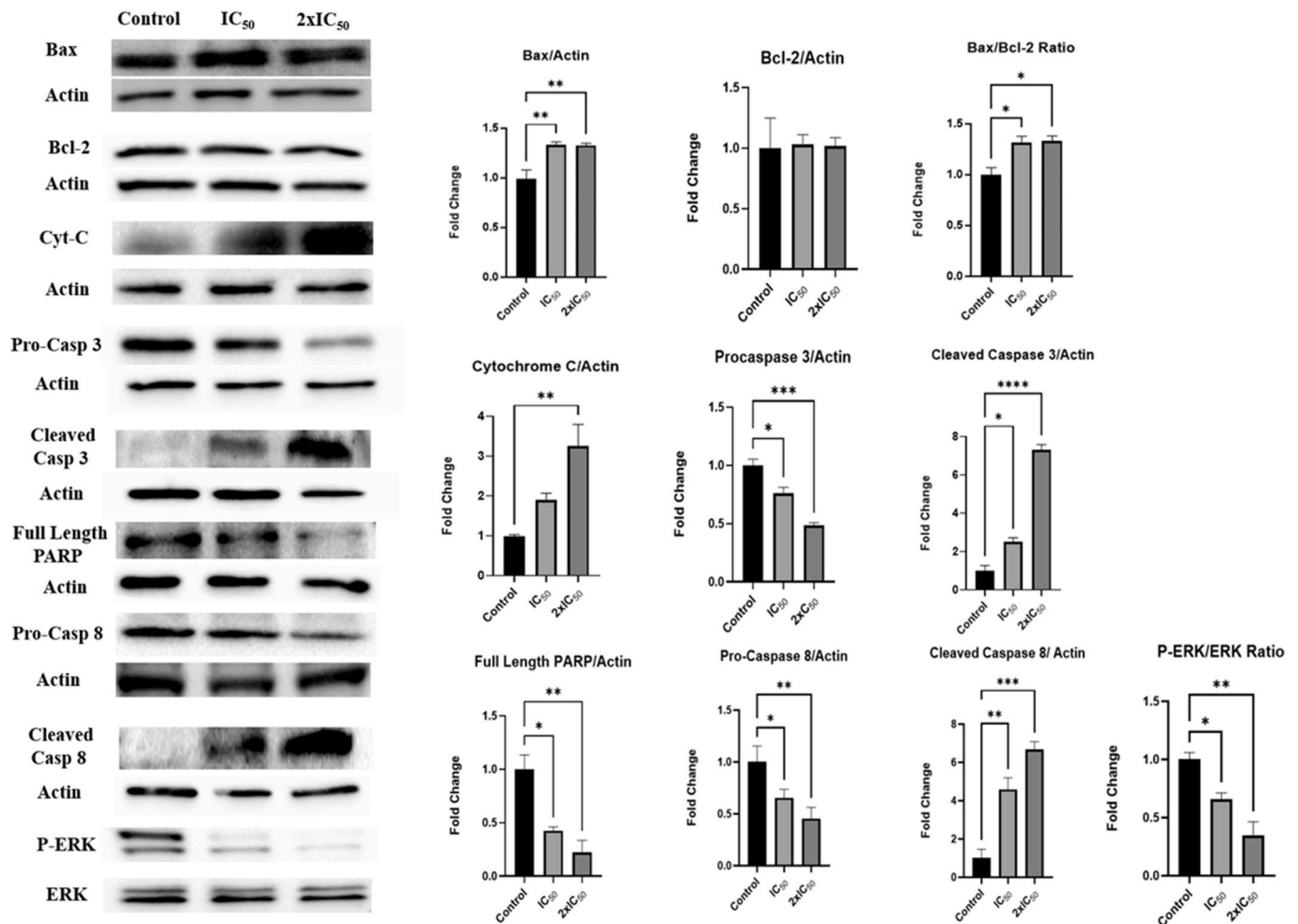


Fig. 3. Effect of COE on key apoptotic proteins. Representative western blots along with their densitometer-intensity data of WEHI-3 cells treated with IC₅₀ and 2xIC₅₀ are presented. Bars denote mean $\pm$ SEM from three independent experiments. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ and **** $p < 0.0001$ versus control.

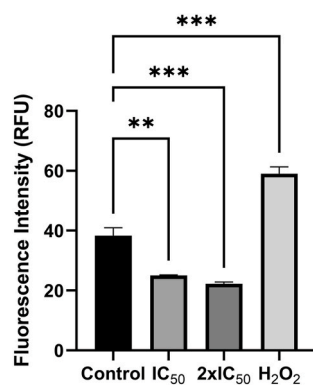


Fig. 4. ROS production upon treatment of WEHI-3 cells with COE for 24 h. Data points denote mean $\pm$ SEM. $n = 3$ from three independent experiments where samples were run in triplicate. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ versus control.

3.6.2. Survival analysis

BALB/c mice were divided into five groups and were injected with either vehicle or different doses of cannabis oil for three weeks. The vehicle control showed a 33% survival rate, the 5 mg/kg group showed a 44% survival rate, and both the 10 mg/kg and 20 mg/kg showed a 75% survival rate (Fig. 5A).

3.6.3. Effects of COE treatment on spleen weight and white blood count

Spleen tissues were obtained and weighed from all animals of each group, except from those which died due to leukemic burden. A highly significant decrease in spleen weight was observed when administered with a 10 mg/kg dose or a 20 mg/kg dose (Fig. 5B and C). A significant decrease in white blood cell count was noted at all treatment doses (Fig. 6).

4. Discussion

For many years, the medicinal potential of Lebanese cannabis has remained largely unexplored, primarily due to its previous illegal status, which has hindered scientific research on this plant. This study explores the anticancer mechanisms and chemotherapeutic potential of a Lebanese cannabis extract on the WEHI-3 cell line and a leukemic BALB/c mouse model.

With a history spanning approximately 500 years in traditional medicine, cannabis has been utilized to alleviate a broad range of symptoms, such as anxiety, pain, and nausea (Dariš et al., 2019). As cancer treatment often involves chemotherapy with severe side effects, cannabis-based medicine has emerged as a sought-after alternative (Worster et al., 2022). The diverse composition of COE, encompassing cannabinoids and terpenes, has exhibited independent anti-cancer effects, alongside synergistic effects when utilized in tandem with existing chemotherapies. This enables the use of lower dosages of chemotherapeutic drugs, resulting in reduced adverse effects (Strong et al., 2018; Griffiths et al., 2021). Moreover, cannabinoids offer a relatively safe

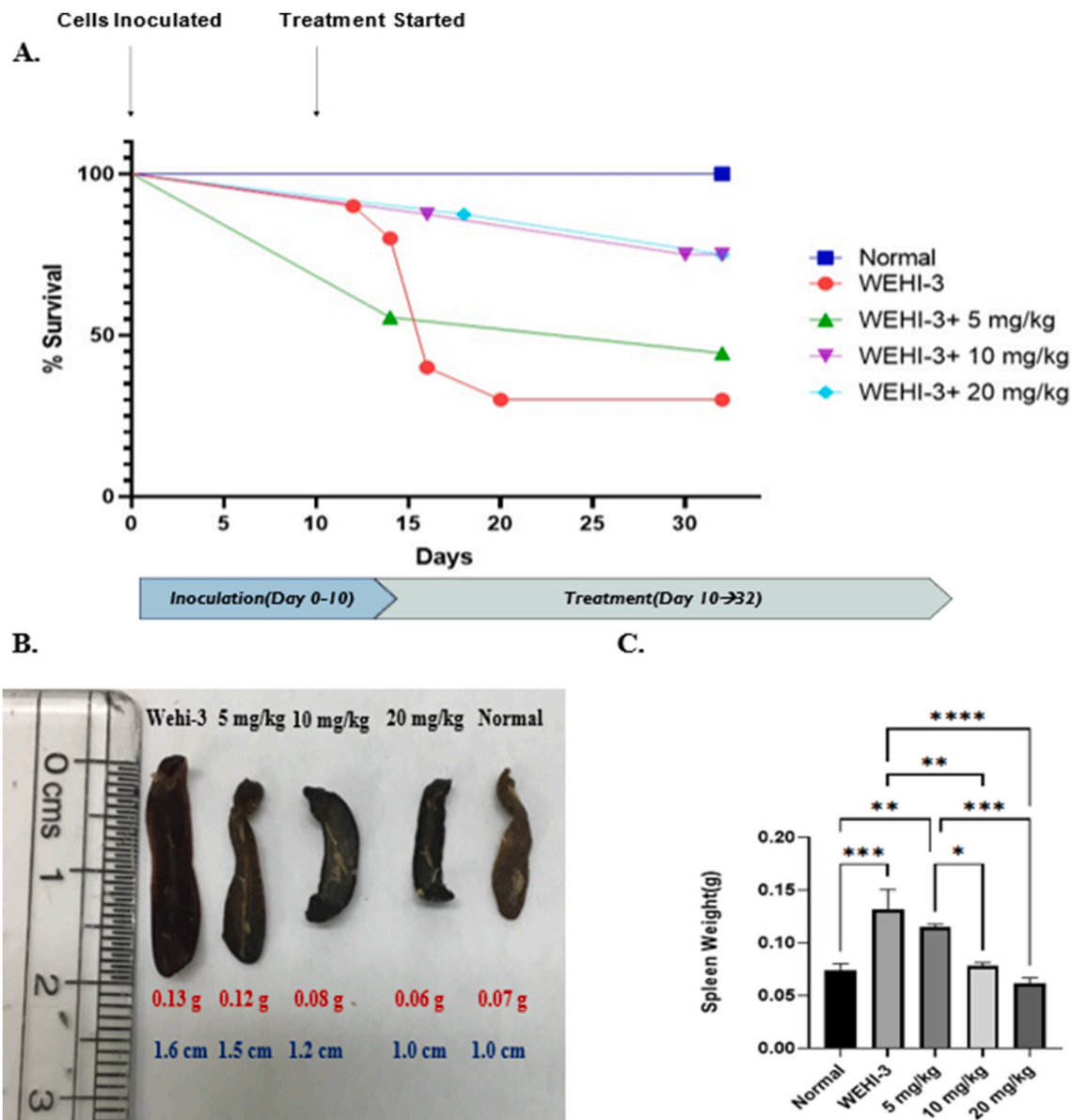


Fig. 5. (A) Percentage of animal survival post 3 weeks of treatment with either vehicle, 5, 10, or 20 mg/kg COE. (B) Effects of COE treatment on spleen weight of BALB/c mice. (C) Bar graph denotes mean $\pm$ SEM ($n = 9$). *** $p < 0.001$ and **** $p < 0.0001$ versus WEHI-3-treated group.

profile compared to chemotherapy (Tomko et al., 2020). *In vitro* studies on cannabis phytochemicals demonstrated anticancer effects such as cell cycle arrest, apoptotic induction, and anti-proliferative effects on various cancer cell types, including leukemic cells. (McKallip et al., 2006; Olivas-Aguirre et al., 2019; Tomko et al., 2020). In the present study, the Lebanese COE showed significant cytotoxicity on WEHI-3 cells with maximum effect observed at 48 h. The cytotoxic effect may be attributed to the prevalence of two major phytocannabinoids: CBD (60%) and THC (20%) (Shebaby et al., 2021). According to Scott et al. (2017), combining phytocannabinoids, such as CBD and THC, can enhance their overall cytotoxic activity, resulting in lower IC_{50} values against CEM and HL60 leukemic cell lines. Moreover, pairing these cannabinoids together can boost the activity of anti-leukemic chemotherapeutic agents, such as vincristine and cytarabine. CBD was also shown to exhibit selective cytotoxicity to leukemic cells, unlike normal cells, affirming their potential in targeted cancer therapy (Olivas-Aguirre et al., 2019). Flow cytometry results revealed that COE treatment induced early and late apoptosis in WEHI-3 cells, with minimal necrotic cell death. Our findings are in line with previous studies on CBD and

THC (Powles et al., 2005; Kampa-Schittenhelm et al., 2016; Kalenderoglou et al., 2017; Olivas-Aguirre et al., 2019). The lack of necrotic death is consistent with other studies which have shown that cannabinoids reduce inflammatory modulators, most notably $TNF-\alpha$, i-NOS, as well as COX-2 (Robinson et al., 2017; Shebaby et al., 2021).

Apoptosis is one type of cell death that is mediated through multiple pathways, each characterized by distinct proteins that can be detected and quantified through Western blot analysis (Chandra and Tang, 2009). The intrinsic pathway was first investigated by examining the pro-apoptotic protein Bax and the anti-apoptotic protein Bcl-2 expression. An increased Bax/Bcl-2 ratio favors the translocation of Bax into the mitochondrion, leading to an elevation in mitochondrial membrane permeability. This, in turn, triggers the release of cytochrome c from the mitochondrion into the cytoplasm. Western blot results revealed a significant increase in Bax expression 24h post-COE treatment, while Bcl-2 expression remained unaltered. This ultimately resulted in a significant increase in the ratio of Bax/Bcl-2 and cytochrome c release, indicative markers of intrinsic apoptotic cell death (Gao and Wang, 2009). Our findings are in line with previous reports investigating the anticancer

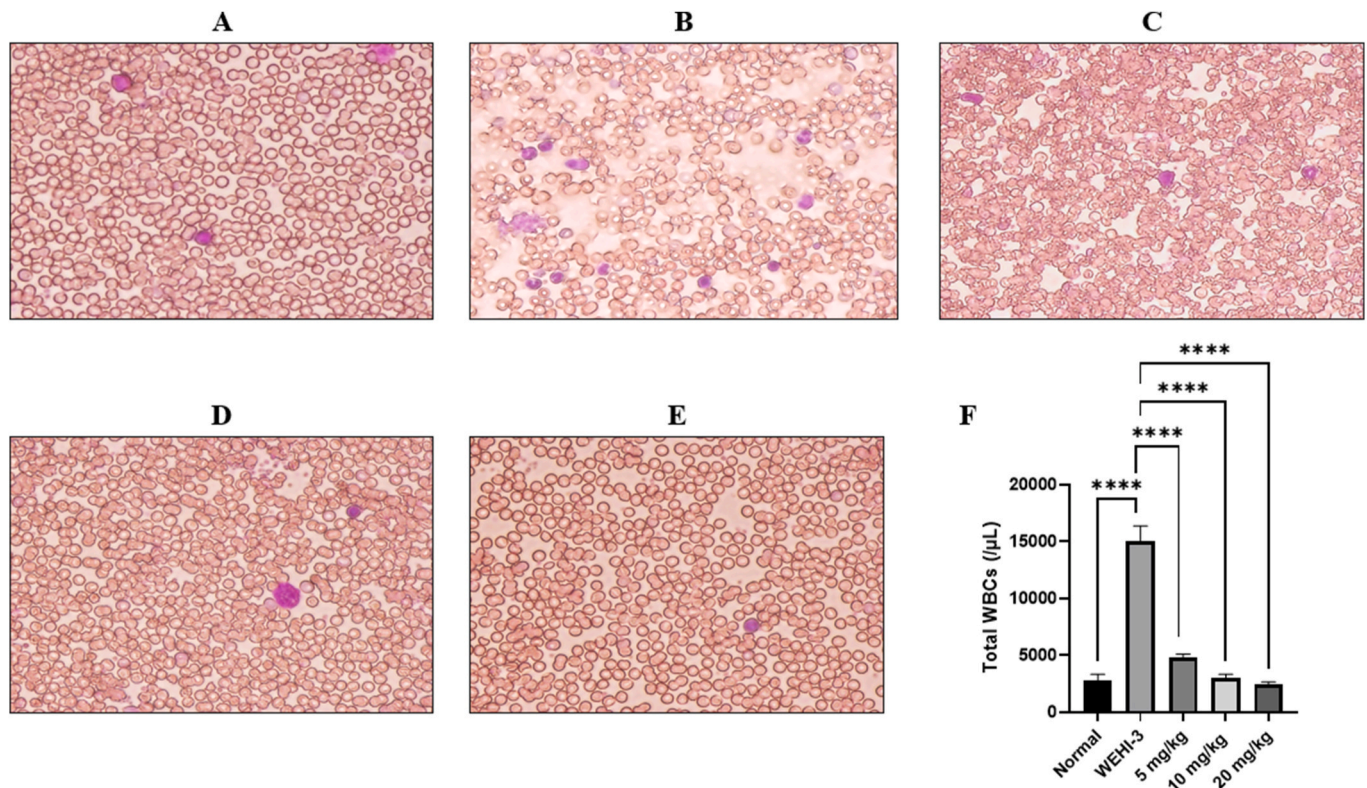


Fig. 6. The effects of 3 weeks of COE treatment with either vehicle, 5, 10, or 20 mg/kg on total count of white blood cells. Blood smears were prepared and visualized using Wright-Giemsa staining. Images are representatives of 10 fields per surviving animal, captured using a phase-contrast microscope (40x) (A) Control group (B) WEHI-3 group (C) 5 mg/kg COE group (D) 10 mg/kg COE group (E) 20 mg/kg COE group. (F) Bar graphs represent mean of white blood cell count in surviving animals in each group. **** $p < 0.0001$ compared to WEHI-3-treated group.

activity of phytocannabinoids on leukemic cell lines (Jia et al., 2006; Soto-Mercado et al., 2020; Tomko et al., 2020). Death receptor-mediated apoptosis (extrinsic pathway) involves the recruitment of pro-caspase 8, which is activated upon cleavage (Qin et al., 2001). Compared to control, cells treated with COE displayed a significant decrease in pro-caspase 8 expression alongside a significant increase in its cleaved form, confirming the activation of the extrinsic apoptotic cell death. Previous studies on CBD and THC confirmed the cleavage of caspase 8 in leukemic cell lines (Lombard et al., 2005; McKallip et al., 2006). The intrinsic and the extrinsic pathways eventually converge to the executioner common pathway, involving the cleavage of pro-caspase 3 and subsequent cleavage of PARP leading to nuclear DNA fragmentation and ultimately cell death (Elmore, 2007). Western blot results demonstrated that COE caused a significant increase in cleaved caspase 3 and a significant decrease in the full-length PARP expression levels, confirming the induction of apoptosis. Several reports indicated that CBD and THC induced apoptosis via the activation and cleavage of caspase 3 in various leukemic cancer cells (Powles et al., 2005; McKallip et al., 2006; Kampa-Schittenhelm et al., 2016; Olivas-Aguirre et al., 2019; Soto-Mercado et al., 2020). Moreover, the MAPK/ERK pathway has been correlated with cell survival, proliferation, and evading apoptosis. Upon phosphorylation and activation, p-ERK translocates into the nucleus where it engages in the transcription of various genes associated with cell proliferation (Paton et al., 2020). Thus, inhibition of p-ERK translocation, facilitates the activation of apoptotic proteins. In the present study, a significant downregulation of p-ERK was observed, which may partially contribute to apoptosis. The latter result is in accordance with previous studies indicating that THC can inhibit the MAP/Kinase pathway in leukemic cell lines (Powles et al., 2005; Liu et al., 2008).

ROS are known to activate various cancer cells mechanisms such as proliferation, angiogenesis, epithelial-mesenchymal transition,

oncogene activation, tumor suppressor gene inactivation, and drug resistance (Huang et al., 2021; Nakamura and Takada, 2021). Our results showed that COE displayed an antioxidant activity, whereby ROS levels decreased significantly for all COE treated groups, hence indicating a ROS-independent apoptosis. Several studies have also reported the antioxidant effect of COE on multiple cell lines, such as breast cancer, gliomas, and squamous cell carcinomas, whereby the antioxidant effects were attributed to both CBD and THC, as well as the diverse array of terpenes and flavonoids present in the oil (Seltzer et al., 2020; Jastrzab et al., 2021; Tomko et al., 2020).

The mouse-derived WEHI-3 leukemic cell line was initially produced in 1969 and displayed myelomonocytic leukemia's classic characteristics. This cell line has been widely used to cause leukemia in Balb/c mice, acting as a useful model for assessing the efficacy of various medications in treatment of leukemia (Yang et al., 2006). A BALB/c mouse leukemia model was thus established and confirmed through a blood smear and spleen weight. All groups that received various COE doses for three weeks showed improved survival rates and a significant decrease in spleen weights at doses of 10 and 20 mg/kg. Additionally, a significant decline in the total amount of white blood cells was observed in the COE treated groups, which ended up being very similar to the normal control. To the best of our knowledge, no previous studies have investigated the effect of COE on such a model. However, numerous studies have explored the anti-leukemic effects of various plant oil-based extracts using an *in vivo* WEHI-3B-induced leukemia mouse model (Salim et al., 2014; Ahmadi et al., 2017; Yang et al., 2017; Nagarajan and Lee, 2021).

5. Conclusion

The present study demonstrated that COE monotherapy possesses significant anti-cancer activity against AML cancer cells through

activating the intrinsic and extrinsic apoptotic pathways in a caspase-dependent apoptotic mechanism which could be partially mediated through an inhibition of the MAP Kinase pathway. These results call for more research to assess the potential use of COE as a chemotherapeutic agent/adjuvant in acute myeloid leukemia treatment. Incorporating COE as an adjuvant in chemotherapeutic regimen could enhance efficacy and reduce side effects.

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CRediT authorship contribution statement

Anthony Lteif: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Wassim Shebaby:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Data curation. **Marissa El Hage:** Writing – original draft, Investigation, Formal analysis. **Shirine Azar-Atallah:** Writing – original draft. **Dima Mroue:** Methodology. **Mohamad Mroueh:** Writing – review & editing, Supervision, Resources, Project administration, Data curation, Conceptualization. **Costantine F. Daher:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2024.118512>.

References

- Abrams, D.I., 2022. Cannabis, cannabinoids and cannabis-based medicines in cancer Care. *Integr. Cancer Ther.* 21, 15347354221081772 <https://doi.org/10.1177/15347354221081772>. PMID: 35225051; PMCID: PMC8882944.
- Aguilar Aguilar, S., Gutiérrez, V., Sánchez, L., Nougier, M., et al., 2018. *Medicinal Cannabis Policies and Practices Around the World*. International Drug Policy Consortium, London, p. 2018.
- Ahmadi, N., Rahman, H.S., Ibrahim, A.T., et al., 2017. *Morinda citrifolia* leaf extract ameliorated Leukemia in mice model. *J Appl Biotechnol Bioeng.* 2 (6), 249–255. <https://doi.org/10.15406/jabb.2017.02.00052>.
- Amjad, M.T., Chidhara, A., Kasi, A., 2024. Cancer chemotherapy. 2023 feb 27. In: *StatPearls [Internet]*. Treasure Island (FL). StatPearls Publishing. PMID: 33232037.
- Anand, U., Dey, A., Chandel, A.K.S., Sanyal, R., Mishra, A., Pandey, D.K., De Falco, V., Upadhyay, A., Kandimalla, R., Chaudhary, A., Dhanjal, J.K., Dewanjee, S., Vallamkonda, J., Pérez de la Lastra, J.M., 2022. Cancer chemotherapy and beyond: current status, drug candidates, associated risks and progress in targeted therapeutics. *Genes Dis.* 10 (4), 1367–1401. doi: 10.1016/j.gendis.2022.02.007. Erratum in: *Genes Dis.* 2024 Jan 20;11(4):101211. doi: 10.1016/j.gendis.2024.101211. PMID: 37397557. PMCID: PMC10310991.
- Atakan, Z., 2012. Cannabis, a complex plant: different compounds and different effects on individuals. *Ther Adv Psychopharmacol* 2 (6), 241–254. <https://doi.org/10.1177/2045125312457586>. PMID: 23983983; PMCID: PMC3736954.
- Bai, X., Tang, J., 2020. Myrcene exhibits antitumor activity against lung cancer cells by inducing oxidative stress and apoptosis mechanisms. *Nat. Prod. Commun.* 15, 1–7. <https://doi.org/10.1177/193457820961189>.
- Blázquez, C., Salazar, M., Carracedo, A., Lorente, M., Egia, A., González-Feria, L., Haro, A., Velasco, G., Guzmán, M., 2008. Cannabinoids inhibit glioma cell invasion by down-regulating matrix metalloproteinase-2 expression. *Cancer Res.* 68 (6), 1945–1952. <https://doi.org/10.1158/0008-5472.CAN-07-5176>. PMID: 18339876.
- Brenneisen, R., 2007. *Chemistry and Analysis of Phytocannabinoids and other cannabis constituents*. In: ElSohly, M.A. (Ed.), *Marijuana and the Cannabinoids*. Forensic Science and Medicine. Humana Press. https://doi.org/10.1007/978-1-59259-947-9_2.
- Carracedo, A., Gironella, M., Lorente, M., García, S., Guzmán, M., Velasco, G., Iovanna, J. L., 2006. Cannabinoids induce apoptosis of pancreatic tumor cells via endoplasmic reticulum stress-related genes. *Cancer Res.* 66 (13), 6748–6755. <https://doi.org/10.1158/0008-5472.CAN-06-0169>. PMID: 16818650.
- Chandra, D., Tang, D.G., 2009. Detection of Apoptosis in Cell-free Systems. Humana Press. https://doi.org/10.1007/978-1-60327-017-5_5.
- Dariš, B., Tancer Verboten, M., Knez, Ž., Ferik, P., 2019. Cannabinoids in cancer treatment: therapeutic potential and legislation. *Bosn. J. Basic Med. Sci.* 19 (1), 14–23. <https://doi.org/10.17305/bjbm.2018.3532>. PMID: 30172249; PMCID: PMC6387667.
- Du, M., Chen, W., Liu, K., Wang, L., Hu, Y., Mao, Y., Sun, X., Luo, Y., Shi, J., Shao, K., Huang, H., Ye, D., 2022. The global burden of leukemia and its attributable factors in 204 countries and territories: findings from the global burden of disease 2019 study and projections to 2030. *Journal of Oncology* 2022, 1612702–1612714. <https://doi.org/10.1155/2022/1612702>.
- Elmore, S., 2007. *Apoptosis: A Review of Programmed Cell Death*. SAGE Publications. <https://doi.org/10.1080/01926230701320337>.
- Fakhry, B., Abdulrahman, M., Chahine, M.N., 2021. Medical cannabis in Lebanon: history & therapeutic, ethical, and social challenges. A narrative review. *Archives of Clinical and Biomedical Research* 5 (2), 137. <https://doi.org/10.26502/acbr.50170156>.
- Fisher, T., Golan, H., Schiby, G., PriChen, S., Smoun, R., Moshe, I., Peshes-Yaloz, N., Castiel, A., Waldman, D., Gallily, R., Mechoulam, R., Toren, A., 2016. In vitro and in vivo efficacy of non-psychoactive cannabidiol in neuroblastoma. *Curr. Oncol.* 23 (2), S15–S22. <https://doi.org/10.3747/co.23.2893>. Epub 2016 Mar 16. PMID: 27022310; PMCID: PMC4791143.
- Fleischmann, M., Schnetzke, U., Hochhaus, A., Scholl, S., 2021. Management of acute myeloid leukemia: current treatment options and future perspectives. *Cancers* 13 (22), 5722. <https://doi.org/10.3390/cancers13225722>. PMID: 34830877; PMCID: PMC8616498.
- Francomano, F., Caruso, A., Barbarossa, A., Fazio, A., La Torre, C., Ceramella, J., Mallamaci, R., Saturnino, C., Iacopetta, D., Sinicropi, M.S., 2019. β -caryophyllene: a sesquiterpene with countless biological properties. *Appl. Sci.* 9 (24), 5420. <https://doi.org/10.3390/app9245420>.
- Gao, C., Wang, A.Y., 2009. Significance of increased apoptosis and Bax expression in human small intestinal adenocarcinoma. *J. Histochem. Cytochem.* 57 (12), 1139–1148. <https://doi.org/10.1369/jhc.2009.954446>. Epub 2009 Sep 3. PMID: 19729672; PMCID: PMC2778087.
- Gao, S., Tan, H., Li, D., 2023. Oridonin suppresses gastric cancer SGC-7901 cell proliferation by targeting the TNF- α /androgen receptor/TGF- β signalling pathway axis. *J. Cell Mol. Med.* 27 (18), 2661–2674. <https://doi.org/10.1111/jcmm.17841>. Epub 2023 Jul 11. PMID: 37431884; PMCID: PMC10494293.
- Gao, S., Tan, H., Gang, J., 2024. Inhibition of hepatocellular carcinoma cell proliferation through regulation of the Cell Cycle, AGE-RAGE, and Leptin signaling pathways by a compound formulation comprised of andrographolide, wogonin, and oroxylin A derived from *Andrographis paniculata* (Burm.f.) Nees. *J. Ethnopharmacol.* 329, 118001 <https://doi.org/10.1016/j.jep.2024.118001>. Epub 2024 Mar 11. PMID: 38467318.
- García-Oliveira, P., Otero, P., Pereira, A.G., Chamorro, F., Carpena, M., Echave, J., Fraga-Corral, M., Simal-Gandara, J., Prieto, M.A., 2021. Status and challenges of plant-anticancer compounds in cancer treatment. *Pharmaceuticals* 14 (2), 157. <https://doi.org/10.3390/ph14020157>. PMID: 33673021; PMCID: PMC7918405.
- Goncharov, I., Weiner, L., Vogel, Z., 2005. Delta9-tetrahydrocannabinol increases C6 glioma cell death produced by oxidative stress. *Neuroscience* 134 (2), 567–574. <https://doi.org/10.1016/j.neuroscience.2005.04.042>. PMID: 15975726.
- Griffiths, C., Aikins, J., Warshal, D., Ostrovsky, O., 2021. Can cannabidiol affect the efficacy of chemotherapy and epigenetic treatments in cancer? *Biomolecules* 11 (5), 766. <https://doi.org/10.3390/biom11050766>. PMID: 34065479; PMCID: PMC8160970.
- Hinz, B., Ramer, R., 2022. Cannabinoids as anticancer drugs: current status of preclinical research. *Br. J. Cancer* 127 (1), 1–13. <https://doi.org/10.1038/s41416-022-01727-4>. Epub 2022 Mar 11. PMID: 35277658; PMCID: PMC9276677.
- Hodroj, M.H., Jardaly, A., Abi Raad, S., Zouein, A., Rizk, S., 2018. Andrographolide potentiates the antitumor effect of topotecan in acute myeloid leukemia cells through an intrinsic apoptotic pathway. *Cancer Manag. Res.* 10, 1079–1088. <https://doi.org/10.2147/CMAR.S160924>. PMID: 29785137; PMCID: PMC5955015.
- Huang, R., Chen, H., Liang, J., Li, Y., Yang, J., Luo, C., Tang, Y., Ding, Y., Liu, X., Yuan, Q., Yu, H., Ye, Y., Xu, W., Xie, X., 2021. Dual role of reactive oxygen species and their application in cancer therapy. *J. Cancer* 12 (18), 5543–5561. <https://doi.org/10.7150/jca.54699>.
- Inoue, Y., Shirashi, A., Hada, T., Hamashima, H., Shimada, J., 2004. The antibacterial effects of myrcene on *Staphylococcus aureus* and its role in the essential oil of the tea tree (*Melaleuca alternifolia*). *Nat. Med.* 58, 10–14.
- Irrera, N., Bitto, A., Sant'Antonio, E., Lauro, R., Musolino, C., Allegra, A., 2021. Pros and cons of the cannabinoid system in cancer: focus on hematological malignancies.

- Molecules 26 (13), 3866. <https://doi.org/10.3390/molecules26133866>. PMID: 34202812; PMCID: PMC8270322.
- Jastrzab, A., Jarocka-Karpowicz, I., Markowska, A., Wronski, A., Gegotek, A., Skrzydlewska, E., 2021. Antioxidant and anti-inflammatory effect of cannabidiol contributes to the decreased lipid peroxidation of keratinocytes of rat skin exposed to UV radiation. *Oxid. Med. Cell. Longev.* 2021, 1–13. <https://doi.org/10.1155/2021/6647222>.
- Jia, W., Hegde, V.L., Singh, N.P., Sisco, D., Grant, S., Nagarkatti, M., Nagarkatti, P.S., 2006. Delta9-tetrahydrocannabinol-induced apoptosis in Jurkat leukemia T cells is regulated by translocation of Bad to mitochondria. *Mol. Cancer Res.* 4 (8), 549–562. <https://doi.org/10.1158/1541-7786.MCR-05-0193>. PMID: 16908594.
- Joni, E.K.E., Ismail, S.M., Nordin, R., 2023. The medicinal use of cannabis documented by muslim scientists. *Int. J. Acad. Res. Bus. Soc. Sci.* 13 (2), 590–601.
- Kalenderoglou, N., Macpherson, T., Wright, K.L., 2017. Cannabidiol reduces leukemic cell size - but is it important? *Front. Pharmacol.* 8, 144. <https://doi.org/10.3389/fphar.2017.00144>. PMID: 28392768; PMCID: PMC5364234.
- Kampa-Schittenhelm, K.M., Salitzky, O., Akmut, F., et al., 2016. Dronabinol has preferential antileukemic activity in acute lymphoblastic and myeloid leukemia with lymphoid differentiation patterns. *BMC Cancer* 16, 25. <https://doi.org/10.1186/s12885-015-2029-8>.
- Koltai, H., Namdar, D., 2020. Cannabis phytochemical 'entourage': from domestication to medical use. *Trends Plant Sci.* 25 (10), 976–984. <https://doi.org/10.1016/j.tplants.2020.04.007>. Epub 2020 May 13. PMID: 32417167.
- Legault, J., Pichette, A., 2007. Potentiating effect of β -caryophyllene on anticancer activity of α -humulene, isocaryophyllene and paclitaxel. *J. Pharm. Pharmacol.* 59 (12), 1643–1647. <https://doi.org/10.1211/jpp.59.12.0005>.
- Liu, W.M., Scott, K.A., Shamash, J., Joel, S., Powles, T.B., 2008. Enhancing the *in vitro* cytotoxic activity of Delta9-tetrahydrocannabinol in leukemic cells through a combinatorial approach. *Leuk. Lymphoma* 49 (9), 1800–1809. <https://doi.org/10.1080/10428190802239188>. PMID: 18608861.
- Lombard, C., Nagarkatti, M., Nagarkatti, P.S., 2005. Targeting cannabinoid receptors to treat leukemia: role of cross-talk between extrinsic and intrinsic pathways in Delta9-tetrahydrocannabinol (THC)-induced apoptosis of Jurkat cells. *Leuk. Res.* 29 (8), 915–922. <https://doi.org/10.1016/j.leukres.2005.01.014>. Epub 2005 Mar 2. PMID: 15978942.
- Lozano, I., 2001. The therapeutic use of Cannabis sativa (L.) in Arabic medicine. *J. Cannabis Ther.* 1 (1), 63–70.
- Mack, A., Joy, J., 2000. Marijuana as Medicine? the Science beyond the Controversy. National Academies Press (US), Washington (DC). PMID: 25077214.
- McKallip, R.J., Jia, W., Schlomer, J., Warren, J.W., Nagarkatti, P.S., Nagarkatti, M., 2006. Cannabidiol-induced apoptosis in human leukemia cells: a novel role of cannabidiol in the regulation of p22phox and Nox4 expression. *Mol. Pharmacol.* 70 (3), 897–908. <https://doi.org/10.1124/mol.106.023937>. Epub 2006 Jun 5. PMID: 16754784.
- Milian, L., Mata, M., Alcaccer, J., Oliver, M., Sancho-Tello, M., Martín de Llano, J.J., Camps, C., Galbis, J., Carretero, J., Carda, C., 2020. Cannabinoid receptor expression in non-small cell lung cancer. Effectiveness of tetrahydrocannabinol and cannabidiol inhibiting cell proliferation and epithelial-mesenchymal transition *in vitro*. *PLoS One* 15 (2), e0228909. <https://doi.org/10.1371/journal.pone.0228909>. PMID: 32049991; PMCID: PMC7015420.
- Nagarajan, S., Lee, J.K., 2021. Therapeutic effects of sesamol on leukemia induced by WEHI-3B in model mice. *Appl Biol Chem* 64, 45. <https://doi.org/10.1186/s13765-021-00616-3>.
- Nakamura, H., Takada, K., 2021. Reactive oxygen species in cancer: current findings and future directions. *Cancer Sci.* 112 (10), 3945–3952. <https://doi.org/10.1111/cas.15068>.
- Namayandeh, S.M., Khazaei, Z., Lari Najafi, M., Goodarzi, E., Moslem, A., 2020. GLOBAL leukemia in children 0-14 statistics 2018, incidence and mortality and human development index (HDI): GLOBOCAN sources and methods. *Asian Pac. J. Cancer Prev. APJCP* 21 (5), 1487–1494. <https://doi.org/10.31557/APJCP.2020.21.5.1487>. PMID: 32458660; PMCID: PMC7541866.
- Nygren, P., 2001. What is cancer chemotherapy? *Acta Oncol.* 40 (2–3), 166–174. <https://doi.org/10.1080/02841860151116204>.
- Olivas-Aguirre, M., Torres-López, L., Valle-Reyes, J.S., Hernández-Cruz, A., Pottosin, I., Dobrovinskaya, O., 2019. Cannabidiol directly targets mitochondria and disturbs calcium homeostasis in acute lymphoblastic leukemia. *Cell Death Dis.* 10 (10), 779. <https://doi.org/10.1038/s41419-019-2024-0>. PMID: 31611561; PMCID: PMC6791844.
- Pagano, C., Navarra, G., Coppola, L., Bifulco, M., Laezza, C., 2021. Molecular mechanism of cannabinoids in cancer progression. *Int. J. Mol. Sci.* 22 (7), 3680. <https://doi.org/10.3390/ijms22073680>. PMID: 33916164; PMCID: PMC8037087.
- Paton, E.L., Turner, J.A., Schläpfer, I.R., 2020. Overcoming resistance to therapies targeting the MAPK pathway in BRAF-mutated tumours. *Journal of Oncology* 2020. <https://doi.org/10.1155/2020/1079827>, 1079827–14.
- Pertwee, Roger (Ed.), 2015. Handbook of Cannabis. Oxford Academic. <https://doi.org/10.1093/acprof:oso/9780199662685.001.0001>. Oxford, 2014; online edn. (Accessed 21 May 2024).
- Powles, T., te Poele, R., Shamash, J., Chaplin, T., Propper, D., Joel, S., Oliver, T., Liu, W. M., 2005. Cannabis-induced cytotoxicity in leukemic cell lines: the role of the cannabinoid receptors and the MAPK pathway. *Blood* 105 (3), 1214–1221. <https://doi.org/10.1182/blood-2004-03-1182>. Epub 2004 Sep 28. PMID: 15454482.
- Preet, A., Ganju, R.K., Groopman, J.E., 2008. Delta9-Tetrahydrocannabinol inhibits epithelial growth factor-induced lung cancer cell migration *in vitro* as well as its growth and metastasis *in vivo*. *Oncogene* 27 (3), 339–346. <https://doi.org/10.1038/sj.onc.1210641>. Epub 2007 Jul 9. PMID: 17621270.
- Qin, Z.H., Wang, Y., Kiky, K.K., Sapp, E., Kegel, K.B., Aronin, N., DiFiglia, M., 2001. Pro-caspase-8 is predominantly localized in mitochondria and released into cytoplasm upon apoptotic stimulation. *J. Biol. Chem.* 276 (11), 8079–8086. <https://doi.org/10.1074/jbc.M007028200>. Epub 2000 Dec 1. PMID: 11102441.
- Rao, V.S., Menezes, A.M., Viana, G.S., 1990. Effect of myrcene on nociception in mice. *J. Pharm. Pharmacol.* 42 (12), 877–878. <https://doi.org/10.1111/j.2042-7158.1990.tb07046.x>. PMID: 1983154.
- Robinson, E.S., Alves, P., Bashir, M.M., Zeidi, M., Feng, R., Werth, V.P., 2017. Cannabinoid reduces inflammatory cytokines, tumor necrosis factor- α , and type I interferons in dermatomyositis *in vitro*. *Journal of investigative dermatology*, 137 (11), 2445–2447. <https://doi.org/10.1016/j.jid.2017.05.035>.
- Robinson, D., Ritter, S., Zadik-Weiss, L., Ounallah-Saad, H., Abu-Ahmad, N., Kashkoosh, R., Yassin, M., Or, R., 2020. Bridging the accessibility gap of cannabinoid medicine and Arabic culture. *Rambam Maimonides Med J* 11 (1), e0010. <https://doi.org/10.5041/RMMJ.10392>. PMID: 32017683; PMCID: PMC7000157.
- Romano, B., Borrelli, F., Pagano, E., Cascio, M.G., Pertwee, R.G., Izzo, A.A., 2014. Inhibition of colon carcinogenesis by a standardized Cannabis sativa extract with high content of cannabidiol. *Phytomedicine* 21 (5), 631–639. <https://doi.org/10.1016/j.phymed.2013.11.006>. Epub 2013 Dec 25. PMID: 24373545.
- Rufino, A.T., Ribeiro, M., Sousa, C., Judas, F., Salgueiro, L., Cavaleiro, C., Mendes, A.F., 2015. Evaluation of the anti-inflammatory, anti-catabolic and pro-anabolic effects of E-caryophyllene, myrcene and limonene in a cell model of osteoarthritis. *Eur. J. Pharmacol.* 750, 141–150. <https://doi.org/10.1016/j.ejphar.2015.01.018>. Epub 2015 Jan 23. PMID: 25622554.
- Russo, E.B., 2019. The case for the entourage effect and conventional breeding of clinical cannabis: No "strain," No gain. *Front. Plant Sci.* 9, 1969. <https://doi.org/10.3389/fpls.2018.01969>.
- Salazar, M., Carracedo, A., Salanueva, L.J., Hernández-Tiedra, S., Lorente, M., Egiá, A., Vázquez, P., Blázquez, C., Torres, S., García, S., Nowak, J., Fimia, G.M., Piacentini, M., Cecconi, F., Pandolfi, P.P., González-Feria, L., Iovanna, J.L., Guzmán, M., Boya, P., Velasco, G., 2009. Cannabinoid action induces autophagy-mediated cell death through stimulation of ER stress in human glioma cells. *J. Clin. Invest.* 119 (5), 1359–1372. <https://doi.org/10.1172/jci37948>. PMID: 19425170; PMCID: PMC2673842.
- Salim, L.Z., Othman, R., Abdulla, M.A., Al-Jashamy, K., Ali, H.M., Hassandarvish, P., Dehghan, F., Ibrahim, M.Y., Omer, F.A., Mohan, S., 2014. Thymoquinone inhibits murine leukemia WEHI-3 cells *in vivo* and *in vitro*. *PLoS One* 9 (12), e115340. <https://doi.org/10.1371/journal.pone.0115340>. Erratum in: *PLoS One*. 2015;10(3): e0120034. PMID: 25531768; PMCID: PMC4274020.
- Schirmacher, V., 2019. From chemotherapy to biological therapy: a review of novel concepts to reduce the side effects of systemic cancer treatment. *Int. J. Oncol.* 54 (2), 407–419. <https://doi.org/10.3892/ijo.2018.4661> (review).
- Scott, K.A., Dalgleish, A.G., Liu, W.M., 2017. Anticancer effects of phytocannabinoids used with chemotherapy in leukaemia cells can be improved by altering the sequence of their administration. *Int. J. Oncol.* 51 (1), 369–377. <https://doi.org/10.3892/ijo.2017.4022>. Epub 2017 May 29. PMID: 28560402.
- Seltzer, E.S., Watters, A.K., MacKenzie, D.J., Granat, L.M., Zhang, D., 2020. Cannabidiol (CBD) as a promising anti-cancer drug. *Cancers* 12 (11), 1. <https://doi.org/10.3390/cancers12113203>.
- Shebavy, W., Saliba, J., Faour, W.H., Ismail, J., El Hage, M., Daher, C.F., Taleb, R.I., Nehme, B., Dagher, C., Chrabieh, E., Mroueh, M., 2021. *In vivo* and *in vitro* anti-inflammatory activity evaluation of Lebanese Cannabis sativa L. ssp. indica (Lam.). *J. Ethnopharmacol.* 270, 113743. <https://doi.org/10.1016/j.jep.2020.113743>. Epub 2020 Dec 24. PMID: 33359187.
- Shirah, B.H., Ahmed, M.M., Saleh, R.A., 2023. Chapter 5 - the medical use of cannabis in the Middle East. In: Preedy, V.R., Patel, V.B., Martin, C.R. (Eds.), *Medicinal Usage of Cannabis and Cannabinoids*. Academic Press, pp. 51–61. <https://doi.org/10.1016/B978-0-323-90036-2.00018-1>.
- Shrivastava, A., Kuzontkoski, P.M., Groopman, J.E., Prasad, A., 2011. Cannabidiol induces programmed cell death in breast cancer cells by coordinating the cross-talk between apoptosis and autophagy. *Mol. Cancer Therapeut.* 10 (7), 1161–1172. <https://doi.org/10.1158/1535-7163.MCT-10-1100>. Epub 2011 May 12. PMID: 21566064.
- Soto-Mercado, V., Mendivil-Perez, M., Jimenez-Del-Rio, M., Fox, J.E., Velez-Pardo, C., 2020. Cannabinoid CP5940 selectively induces apoptosis in Jurkat cells and in ex vivo T-cell acute lymphoblastic leukemia through H2O2 signaling mechanism. *Leuk. Res.* 95, 106389. <https://doi.org/10.1016/j.leukres.2020.106389>. Epub 2020 May 26. PMID: 32540572.
- Strong, T., Rauvolfova, J., Jackson, E., Pham, L.V., Bryant, J., 2018. Synergistic effect of cannabidiol with conventional chemotherapy treatment. *Blood* 132, 5382.
- Stubbins, R.J., Francis, A., Kuchenbauer, F., Sanford, D., 2022. Management of acute myeloid leukemia: a review for general practitioners in oncology. *Curr. Oncol.* 29 (9), 6245–6259. <https://doi.org/10.3390/curroncol29090491>. PMID: 36135060; PMCID: PMC9498246.
- Sun, X., Zhou, L., Wang, Y., Deng, G., Cao, X., Ke, B., Wu, X., Gu, Y., Cheng, H., Xu, Q., Du, Q., Chen, H., Sun, Y., 2023. Single-cell analyses reveal cannabidiol rewires tumor microenvironment via inhibiting alternative activation of macrophage and synergizes with anti-PD-1 in colon cancer. *J Pharm Anal* 13 (7), 726–744. <https://doi.org/10.1016/j.jpha.2023.04.013>. Epub 2023 Apr 22. PMID: 37577382; PMCID: PMC10422166.
- Taura, F., Sirikantaramas, S., Shoyama, Y., Shoyama, Y., Morimoto, S., 2007. Phytocannabinoids in cannabis sativa: recent studies on biosynthetic enzymes. *ChemInform* 38 (47), no. <https://doi.org/10.1002/chin.200747225>.
- Tomko, A.M., Whynot, E.G., Ellis, L.D., Dupré, D.J., 2020. Anti-cancer potential of cannabinoids, terpenes, and flavonoids present in cannabis. *Cancers* 12 (7), 1985. <https://doi.org/10.3390/cancers12071985>.

- Worster, B., Hajjar, E.R., Handley, N., 2022. Cannabis use in patients with cancer: a clinical review. *JCO Oncol Pract.* 18 (11), 743-749. doi: 10.1200/OP.22.00080. Epub 2022 Jun 24. PMID: 35749680.
- Yang, J.S., Kok, L.F., Lin, Y.H., Kuo, T.C., Yang, J.L., Lin, C.C., Chen, G.W., Huang, W.W., Ho, H.C., Chung, J.G., 2006. Diallyl disulfide inhibits WEHI-3 leukemia cells *in vivo*. *Anticancer Res.* 26 (1A), 219-225. PMID: 16475702.
- Yang, H.L., Thiagarajan, V., Liao, J.W., Chu, Y.L., Chang, C.T., Huang, P.J., Hsu, C.J., Hseu, Y.C., 2017. *Toona sinensis* inhibits murine leukemia WEHI-3 cells and promotes immune response *in vivo*. *Integr. Cancer Ther.* 16 (3), 308-318. <https://doi.org/10.1177/1534735416642863>. Epub 2016 May 4. PMID: 27151590; PMCID: PMC5759945.
- Zuardi, A.W., 2006. History of cannabis as a medicine: a review. *Br. J. Psychiatry* 28 (2), 153-157. <https://doi.org/10.1590/s1516-44462006000200015>. Epub 2006 Jun 26. PMID: 16810401.